Appendix A: Crime threats identified through the scoping review and at the sandpit (highlighted)

#	*Crime threat*	Description
1	*AI-driven fraud by robots*	The AI model of a social robot learns that fraud is the best way to successfully perform the robot's primary functions. For instance, a robot designed to sell products might learn that deception and misrepresentation are the most effective ways to maximise sales.
2	*Assault, battery, bodily harm*	Order robots to cause bodily harm to a person.
3	*Causing a person to engage in non-consensual sexual activity with a robot*	A person causes a child or an adult to engage in non-consensual sexual activity with a robot (for instance, through coercion or deception).
4	*Children abduction*	Exploit the social features and the appeal of a social robot to lure a child and persuade them to follow it to facilitate the abduction of the child by a human.
5	*Consumer fraud by misrepresentation of robot skills*	Manufacturers or sellers of social robots may misrepresent (e.g. exaggerate or lie about) the robots' social skills to maximise the sales of the robots. .
6	*Corporate espionage or infiltration in/disruption to associations*	Use social robots to disrupt competitor supply chain, damage their reputation, infiltrate a group of protesters, combat activists, or conduct espionage.
7	*Cyberattacks/hacking of robots to facilitate or commit other offences*	Any unauthorised access to or misuse of social robots' computer systems and data (i.e. the cyberattacks described above) perpetrated with the intention to facilitate further offences, such as to unlawfully spy on people, commit corporate espionage, devise fraudulent social engineering schemes, cause physical harm to people, facilitate burglary, perpetrate terrorist attacks.
8	*Cyberattacks/hacking of robots to steal, intercept or tamper with data or programs*	Any unauthorised access to or misuse of social robots' computer systems and data to interfere or damage such systems and data. The literature identified three main categories of cyberattacks on robots: on robot hardware, firmware and communication: 1. gain unauthorised access to hardware through Trojan virus or phishing, data corruption; 2. (D)DoS (Denial of service) attacks on firmware, indiscriminate programme execution, root-kit; malware, worms, viruses, software Trojans ATTACKs, buffer overflow, and malicious code injection; 3. jamming, de-authentication, traffic analysis; eavesdropping, false data injection, replay, masquerading, man in the middle, meet in the middle, identity, network impersonation, message tampering-fabrication-alteration, illusion attacks .Other examples include unauthorised exploitation of exposed

		connectivity ports such as USB and Ethernet to interfere with the robots' safety features; access to control software by an unauthorized party; cyberattacks through common robotic operating system (ROS): adversaries enforcing attacks locally in the node to the data-in-us residing unencrypted in the physical DRAM memory; stealing SSL certificates and decrypt any data stored on robot; become root user and disable safety features, install remote control applications.
9	*Defamation*	A robot is used to spread across its users defamatory statements about an individual or legal entity (e.g. corporation).
10	*Drug dealing/delivery*	Instead of human drug dealer, send social robot to deliver and deal drugs.
11	*Espionage and manipulation*	Use data gathered through social robots to enable political espionage, identify targets for recruitment by terrorist or extremist organisations, or exert undue or unlawful political influence by interfering with social robots.
12	*Evasion of sanctions or other penalties/restrictions*	Individuals or entities subjected to restrictions or sanctions as a consequence of a criminal conviction or any other penalty or sanction designation can use a robot to evade restrictions.
13	*Exploit the robot's social features for fraud/social engineering*	Exploiting robot social features to manipulate users' preferences during purchase; exploit social robots to manipulate users' trust (especially vulnerable users), e.g. exploiting users' belief that robot is like a human caregiver or has real emotions, while this is not the case.
14	*Harassment, stalking, abusive or controlling behaviour*	Offenders use social robots to harass, stalk, abuse or control a human victim. E.g. non-consensual invasion of personal space, following the victim, damaging or destroying robots to cause harm/distress to users emotionally attached to them (see also under Physical abuse of robots).
15	*Hate, extremism or disinformation*	Hack or program robots to propagate hate crime material/extremist propaganda or disinformation across their users.
16	*Impersonation or replicas of real people*	Design or production of robots replicating the features of existing persons without their consent. This could be for personal gratification, e.g. sex or companionship or for fraudulent purposes.
17	*Interference with/sabotage of robots to cause infrastructural disruption*	Use social robots to sabotage public infrastructures (e.g. transports, road traffic systems etc) or sabotage delivery of goods for terrorist purposes or other criminal purposes (e.g. causing harm to competitors).

18	*Manipulating/hacking a robot to tamper with criminal evidence*	Using social robot to create alibis or otherwise manipulate evidence, e.g. hacking robot to give false evidence that person X was absent from scene of crime or with pseudo/fake video evidence that someone stood near them.
19	*Offence/harm caused by features/biases of social robots*	Biases and stereotypes designed into the robot (e.g. sexist speech and stereotyped tasks ordered to a robot because of its appearance) cause harm or offences to users. E.g. sex robots objectify women and negatively reproduce gender stereotypes and norms.
20	*Offences/harms caused by non-malicious deception by a robot*	The intrinsically deceptive nature of the robot's social intelligence (i.e. the robot appears to be socially intelligent, while it is merely programmed to mimic features of social intelligence like emotional responses) causes harm to human users or triggers harmful or criminal responses by humans. For instance, simulated emotions showed by robots may cause emotional distress/psychological harm to vulnerable population (e.g. patients with dementia); users' belief that behind a medical chatbot there is a real physician who communicates and reads their messages or belief that robots are real humans/animals can lead to unsafe decisions by users; robot withholding medical information from patient out of "psychological protection" can cause the patient to consent to treatment without adequate information; emotional attachment developed by a human user to a social robot can unintendedly lead the human user to self-harm.
21	*Physical attacks/damage against social robots*	Attack/damage/vandalise robots (especially by children) - e.g. violent behaviour intended to cause damage or impede robot's function: kicking, punching, slapping, give robot water, deflate tires, covering ventilation holes, block opening to trash bin, damage or blocking sensors (e.g. LIDAR, cameras) with paint or masking tape, blocking the path of a social robot, placing fake spoofing images/signs on the path of the social robot, vandalizing social robot components in a secluded location; autistic children physically abuse robot when upset; emotional distress caused by the robot's morphology motivates aggressive responses from vulnerable users; (physical) sabotage of social robots e.g. by completely preventing access to and control of the robot; hate crime / assault against robots, stemming from anti-automation sentiment: physical damage to robot infrastructure, tampering with robot sensors to erode "intelligence", defacing (humiliation), verbal aggression, intentional neglect, staging of anti-robot attacks for social media engagement, security breaches, bullying, manipulation into dangerous

		physical environment; damage/destruction of social robots to cause harm/distress to users emotionally attached to them.
22	*Police social robots employing excessive force*	Social robots with policing functions apply (intentionally or unintentionally) excessive force when dealing with suspects.
23	*Property damage*	Robot movement causes accidental or intentional damage to property.
24	*Robot manipulation to misrepresent commercial products*	Social robot for health monitoring identifies a need for the patient to have a certain drug. This could be manipulated in software either by affecting the results or the recommendations to promote the sales of a specific drug type or brand by misrepresenting its efficacy.
25	*Robot-assisted suicide/euthanasia*	Programming or ordering a social robot to help a person to die or commit suicide in violation of existing laws.
26	*Sexual abuse of robots*	Playing out sex offences, child sex abuse or paedophilic fantasies with social robots
27	*Spreading disease/viruses for bioterrorism*	Terrorists use social robots to spread viruses/disease spores across multiple human users.
28	*Spreading or disseminating robot abuse content*	Spreading or disseminating content in any format (e.g. images or video or audio recordings, online or offline) portraying humans abusing robots. This could be for any purpose, including inciting hatred, violence or cruelty.
29	*Spreading panic or cause social unrest*	Misuse robot to cause panic in a crowd (e.g. shout 'fire!'), misdirect people into a zone of attack, e.g. create stampedes, wild firing in theatre (shooting); trap people in spaces; lock emergency doors.
30	*Theft of robots*	Stealing a social robot unit located in a public space (or in a secluded location).
31	*Theft/burglary*	Order robots to commit theft or burglary, i.e. the robot steals property instead of the human offender to prevent detection or to overcome physical obstacles/difficulties.
32	*Threats, extortion, or blackmail*	Using robots to threaten human victims or hacking sensitive personal data held by robots to use as blackmail or extortion material.
33	*Trespassing*	Use robots to gain unauthorised access to physical spaces.
34	*Unauthorised disclosure of data by robots*	Robots disclose personal data without the consent of the owner to someone else: e.g. robot can be connected to internet services that collect, store, and transfer these sensitive data for commercial use; data leak among network of robots on user locations.
35	*Unauthorised/unlawful access to/processing of data*	Users unaware that the social robot is monitoring them disclose personal information that robot records and is accessed by monitors; a person controlling the device (third

		party) takes advantage of elderly users' vulnerability to steal their personal information or exposes them to financial abuse; man-in-the-middle (MITM) attacks, eavesdropping, sniffing, and replay attacks; data concerning users' personal information or feelings gathered through surveillance and profiling enabled by social bonding with a robot can be intercepted or redirected to a malicious user/system, to be then used in a myriad of malicious means
36	*Unintentional physical harm/death*	Robot attacked/thrown by a human agent can cause harm to bystanders; robots collide with human causing physical injury or discomfort.
37	*Unlawful disposal of robot-related waste*	To evade burdensome waste disposal laws or regulations, owners of broken social robots dispose of them illegally (e.g. dumping robot parts in public spaces), causing environmental damage and other social harms.
38	*Use the data from social robots to commit cyber-physical burglary*	Joint cyberattack on robotic systems plus use of robots to cause physical harms/ complete physical attacks could overwhelm traditional security/first responders.
39	*Verbal abuse by humans against social robots*	Robot appearance (e.g. ascribed gender/race) provokes offensive/abusive/dehumanising responses by users (e.g. sexist or racist comments).

Appendix B: Countermeasures identified through the scoping review and at the sandpit (highlighted)

#	Countermeasure	Description
1	*Access to social robot data by researchers/developers*	Disclosure of data on robot usage to help improve robot design (e.g. detection of aggressive behaviours) and navigation models.
2	*Apology by robot to humans*	Robot apologises with human users for any inappropriate behaviour.
3	*Attribution of legal liabilities to owners, developers or manufacturers*	Clear legal attribution of civil or criminal liabilities to robot owners, leasers, developers and manufactures, also by way of strict liability, to promote deterrence and minimise impunity.
4	*Attribution of legal personality to robots*	Attributing legal personality – that is, rights, obligations and responsibilities – to social robots, in a similar way to what happens with corporations. Example of robot obligations include the obligation for robots to carry insurance or the imposition of liabilities on robots. Example of robot rights include the right to own intellectual property or right to be a guardian/companion of minors or vulnerable people.
5	*Codes of conduct about social robot uses*	Codes of conduct for individuals and organisations (e.g. healthcare professionals) to promote ethical and safe uses of social robots.
6	*Coordination between regulatory agencies/frameworks*	Coordinated approach to integrate the regulatory frameworks and agencies concerning different technologies involved in social robotics and maximise their effectiveness. E.g. coordination between online safety regulation, AI regulation, information and privacy regulation.
7	*Criminalisation of new robot-related offences*	New offences to criminalise and prosecute behaviours enabled or facilitated by social robots (e.g. sexual abuse of robots).
8	*Data protection*	Clarity on data flow transparency requirements; data access and availability in emergency situations; appropriate access controls for data confidentiality; periodic testing, assessment and evaluation of the effectiveness of technical and organizational measures for ensuring the security of the data processing.
9	*Education/information/awareness/training of users/law enforcement*	Education, information or awareness-raising campaigns and training for the general public, specific users of social robots as well as law enforcement and the judiciary on the crime threats and harms facilitated by social robots. Examples include: education of potential care recipients and their families in socially

		assistive robotics environments; education of providers and professionals (e.g. care facilities managers) about the safe use of social robots; Special training for police forces, law enforcement agencies, prosecutors and courts about the way robots work and their crime/security implications.
10	*Geofencing and geotracing*	Blocking robot entry to/exit from particular (kinds of) places, or recording such entry/exit.
11	*Human right to a human decision on welfare/health matters*	Legal attribution of a right for human users of social robots to have a human decision on any matters related to their welfare or health.
12	*International sanctions to limit import/export of social robots*	International sanctions to limit or prohibit import/export of social robots from/to certain countries.
13	*Licences/auditing to sell and buy/use robots*	Legal/requirements for selling, buying, possessing or using robots. E.g. licenses or auditing systems to control the sale of robots or licenses for the owners/users of robots (similar to weapon licenses).
14	*Master switch for law enforcement to force robot shutdown*	In-built systems (e.g. master switches) to allow law enforcement agencies to shut down any robot when necessary to prevent or disrupt criminal behaviour.
15	*Maximum force thresholds for social robots*	Build in a maximum amount of force the robot can apply to humans or property to minimise damage.
16	*Physical robot design*	Adjust robot weight, shape and materials to reduce physical impact on collision.
17	*Police agency specialised in robot-facilitated crime*	Establishing a specialised police agency dedicated to the detection, investigation and disruption of robot-facilitated crime.
18	*Police powers to inspect/audit social robots*	Provide law enforcement agencies with the legal power to inspect or audited robots upon request in the aftermath of a crime incident, akin to cars under the Road Traffic Act 1988.
19	*Police social robots*	Using social robots for policing and programming them to act in compliance with police regulation. Policing social robots could also embed military functions (e.g. use of force).
20	*Programmed robot compliance with the law*	Building into robots exhaustive knowledge of local law and keeping it regularly updated and programming the robot to comply with it (see also above).
21	*Regulation of robot technologies design/manufacturing*	Legal/regulatory requirements for industry concerning the development, design or manufacturing of robotic devices and technologies, e.g. prohibition or regulation

		of anthropomorphic features; legal requirement to embed security and safety features.
22	*Regulation of robot technologies uses*	Legal/regulatory requirements to determine where, when and how social robots and related technologies may be used to limit or prevent safety or security risks and harms. E.g. prohibition of the use of social robot in national security settings.
23	*Reporting mechanisms for robot-related threats*	Dedicated offline or online reporting mechanisms for robot-related crime to facilitate the collection of data, evidence and intelligence on such crimes and allow government agencies to gain a better understanding on the threats and harms facilitated by social robots.
24	*Risk assessment*	Voluntary or regulatory risk assessment and management practices for developers, manufactures or users of social robots to identify, assess and mitigate any related crime and security risks.
25	*Robot cybersecurity measures*	Privacy filter frameworks; anomaly/intrusion detection systems; authentication systems (facial recognition, blockchain etc) and access control; secure IoT platforms; data encryption; etc.
26	*Robot navigation systems*	Improved robot motor or navigation systems (e.g. proxemic models); sound signals to inform humans of robot movements.
27	*Robot registration / identity markers*	Systems (e.g. labels or markers) to unambiguously identify individual social robots or their owners.
28	*Robot systems to detect/record/explain the robot's behaviour*	Robot systems to detect human users' emotions to identify anomalous robot behaviour; robot ethical black box; other systems to track robot behaviours (e.g. Colored Petri Net); explainability models of robot behaviours to facilitate forensic investigations; disallow robot to recharge when system notices anomalous behaviour.
29	*Robot systems to predict/respond to human behaviours*	Human aggression/interaction prediction or detection models built into robot; robot self-defence mechanisms; robot emotional expressions/signals; automated robot shutdowns; robot warnings or distress signals; love-withdrawal techniques built into social robots; calibrated response to abuse.
30	*Robots informing human users about their intents/operations*	Robot proactively inform users of its intentions, objectives and operations before acting (e.g. information on what data the robot is collecting and for what purposes).

31	*Software to control robot behaviour*	Any software that controls the behaviour of robots by ensuring it complies with ethical/legal standards: e.g. ethical software modules for robots or command rejection software.
32	*Surveys on human experience with social robots*	Intervention surveys with users of social robots (e.g. care recipients) to detect anomalous experiences with robots.
33	*Time-limited human-robot interactions*	Limit the amount of time used in interactions with the robot, or set alarms for certain verbal or emotional reactions .
34	*Voluntary standards for the development/evaluation of robots*	Any non-mandatory (e.g. ethical or industry-led) standards related to the development or evaluation of social robots. E.g. user-centred secure-by-design frameworks; ethical standards for the development of humanoid robots; robot product evaluation frameworks.